

Team #23 Project Proposal**Data-Driven Pit Stop Decision Support for Formula 1 Racing****Team Lead: Alec Slim****Team Members: Bryan Ayala-Rivera, Robert Froberg, Anshul Mehta****Introduction**

Formula 1 (F1) racing is a highly data-driven sport in which teams continuously analyze vehicle performance, tyre degradation, and race conditions to optimize strategy. One of the most consequential strategic decisions during a race is determining when to conduct a pit stop. A poorly timed pit stop can cost several seconds relative to competitors and may significantly alter race outcome. Teams must balance tyre degradation, track position, traffic, and safety car conditions when deciding when to change tyres.

The objective of this project is to develop a data-driven decision support system that recommends optimal pit stop timing based on the current state of a car during a race. Using historical race telemetry and tyre performance data, we aim to build an expert recommender system that predicts when a pit stop would maximize performance outcomes. The system will leverage historical Formula 1 race data and machine learning techniques to provide pit strategy recommendations to the racing team.

Dataset Selection

This project will use historical Formula 1 telemetry and race data accessible through the FastF1 Python package (Schäfer, 2024a). FastF1 provides a programmatic interface to official Formula 1 timing and telemetry data, including lap times, tyre usage, pit stop events, weather conditions, and race session information.

FastF1 enables retrieval of detailed race data such as: lap times and sector times, tyre compounds and tyre age, pit stop timing, track position and race order, safety car deployments, and driver and team performance metrics.

The package has been widely used for motorsport analytics and provides structured datasets suitable for large-scale data processing and analysis. An example of tyre strategy analysis using this dataset is provided in the FastF1 documentation (Schäfer, 2024b).

To complement the telemetry data, this project will reference prior research on Formula 1 tyre strategy and pit stop optimization (Sasikumar, Leema, & Balakrishnan, 2024). This research highlights how tyre degradation, track position, and race dynamics influence strategic decision-making during races. Deep learning techniques were used in this analysis to determine

the pitting strategy; our project will leverage traditional machine learning methods to increase explainability of recommendations, lessen the risk of overfitting the model, and reduce computational requirements.

Problem Definition

The core research question for this project is:

Given the current state of a Formula 1 race, when should a team perform a pit stop to optimize race performance?

During a race, teams must determine whether remaining on track with degrading tyres is beneficial compared to the time lost entering the pit lane and switching tyres. This trade-off depends on several dynamic variables, including tyre wear, lap performance trends, race traffic, and safety car events.

This project aims to construct an expert recommendation model that evaluates the current race state and predicts whether a pit stop should occur within the next lap window.

Key variables that may influence pit stop decisions include: current lap time trends, tyre compound type, tyre age (laps since last pit stop), driver position in the race, time gap to nearby competitors, track conditions, and safety car or virtual safety car deployment.

Project Plan

For this project we will use python to explore the data and build our model through libraries and packages such as Pandas, Matplotlib, and Scikit-learn. We plan on using a traditional binary classification model to make the pit predictions where positive cases indicate a predicted pitstop and negative cases indicate the car should stay out on the track. We will use F1 score as our primary metric for analysis due to the imbalance of laps that involve a pit stop vs laps that do not but we will also use other metrics such as accuracy to help evaluate the model. Finally, the project will be uploaded on a github repository for easy version management and file sharing.

References

Sasikumar, A., Leema, A. A., & Balakrishnan, P. (2024). *Data-driven pit stop decision support for Formula 1 using deep learning models*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12626961/>

Schäfer, P. (2024a). *FastF1*. <https://docs.fastf1.dev/>

Schäfer, P. (2024b). *Tyre strategies during a race*. https://docs.fastf1.dev/gen_modules/examples_gallery/results_strategy/plot_strategy.html